

PROGRESSIVE INDUSTRIAL ANALYTICS TO AI/ML TRANSFORMATION CAPSTONE

End-to-End Supply Chain Optimization & Risk Forecasting

Phase 1-Data Analytics

Submitted by

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Acknowledgement

We express our sincere gratitude to **EVOASTRA VENTURES Pvt. Ltd.** for providing us with the opportunity to undertake this internship project and gain practical exposure in the field of Data Analytics and Supply Chain Analysis.

We would like to convey our heartfelt thanks to our mentor and project guide for their valuable guidance, support, and continuous encouragement throughout the completion of **Phase 1 – Data Analytics (Descriptive & Diagnostic)** activities.

We also extend our gratitude to the faculty members of our department and our college for providing the necessary support and resources during the internship period.

Finally, we thank all team members for their cooperation, dedication, and collective effort in successfully completing this phase of the project.

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Abstract

This internship project focuses on **Supply Chain Analytics using Descriptive and Diagnostic Data Analysis**. The primary objective of Phase 1 was to analyze large-scale historical supply chain datasets and derive meaningful business insights from operational data.

The project involved data cleaning and preprocessing activities including handling missing values, removing duplicates, standardizing formats, and preparing datasets containing over 100K records. Exploratory Data Analysis (EDA) techniques were applied to identify patterns, trends, and anomalies across procurement, warehousing, logistics, and inventory operations.

Key Performance Indicators (KPIs) such as Order Fulfillment Rate, Inventory Turnover, Lead Time Variability, Perfect Order Rate, and Cost-to-Serve were analyzed to understand supply chain performance. Interactive dashboards and visual reports were created to present findings effectively.

The outcomes of this phase provide a structured foundation for future predictive modeling and optimization phases of the project.

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INTRODUCTION

Supply chain management plays a crucial role in ensuring the smooth flow of products, information, and services across procurement, warehousing, transportation, inventory, and customer delivery processes. Modern supply chain systems generate large volumes of data from various operational activities, making data analytics essential for extracting valuable business insights.

Descriptive and diagnostic analytics help organizations understand historical performance, identify trends, detect anomalies, and determine the factors influencing operational efficiency. Through effective analysis of supply chain datasets, businesses can improve decision-making and optimize processes.

This internship project focuses on analyzing large-scale supply chain datasets using data analytics techniques. The work performed in **Phase 1** involved data cleaning, exploratory data analysis, KPI identification, dashboard development, and root cause analysis to understand historical patterns and support future predictive phases.

COMPANY PROFILE

EVOASTRA VENTURES Pvt. Ltd.

EVOASTRA VENTURES Pvt. Ltd. is a technology-driven organization focused on providing innovative solutions in areas such as Data Analytics, Artificial Intelligence, Machine Learning, Web Technologies, and industry-oriented project development. The organization provides practical exposure and hands-on learning opportunities through real-world projects and internship programs.

The company emphasizes skill development by allowing students and trainees to work on industry-based challenges and modern technology frameworks. Through structured project phases and mentorship, participants gain experience in problem-solving, analytical thinking, and practical implementation of technologies.

As part of the internship program, the organization assigned a supply chain analytics project consisting of multiple phases. Phase 1 focused on descriptive and diagnostic analysis of large-scale supply chain datasets to identify trends, define key performance indicators, and derive business insights.

OBJECTIVES

The objective of **Phase 1 – Data Analytics (Descriptive & Diagnostic)** is to analyze large-scale supply chain datasets and extract meaningful insights from historical operational data. This phase focuses on understanding supply chain information generated across procurement, warehousing, inventory management, logistics, and order processing systems. Since large datasets often contain inconsistencies and unstructured information, proper preprocessing and analysis are necessary to support effective business decisions.

The phase begins with data cleaning and preparation activities involving datasets containing more than 100K records. Data preprocessing includes handling missing values, removing duplicate records, identifying outliers, and standardizing formats to improve data quality and reliability. Exploratory Data Analysis (EDA) is performed to identify trends, patterns, seasonal behavior, and operational anomalies within the supply chain process.

The specific objectives of this phase include:

- To analyze large-scale supply chain datasets and preprocess historical data.
- To perform data cleaning activities including handling missing values and duplicates.
- To conduct Exploratory Data Analysis (EDA) to identify trends and patterns.
- To define and analyze supply chain Key Performance Indicators (KPIs).
- To build interactive dashboards and generate business insights.
- To perform root cause analysis and support future project phases.

The completion of these objectives provides a strong foundation for upcoming phases involving predictive analysis and advanced decision-making models.

METHODOLOGY

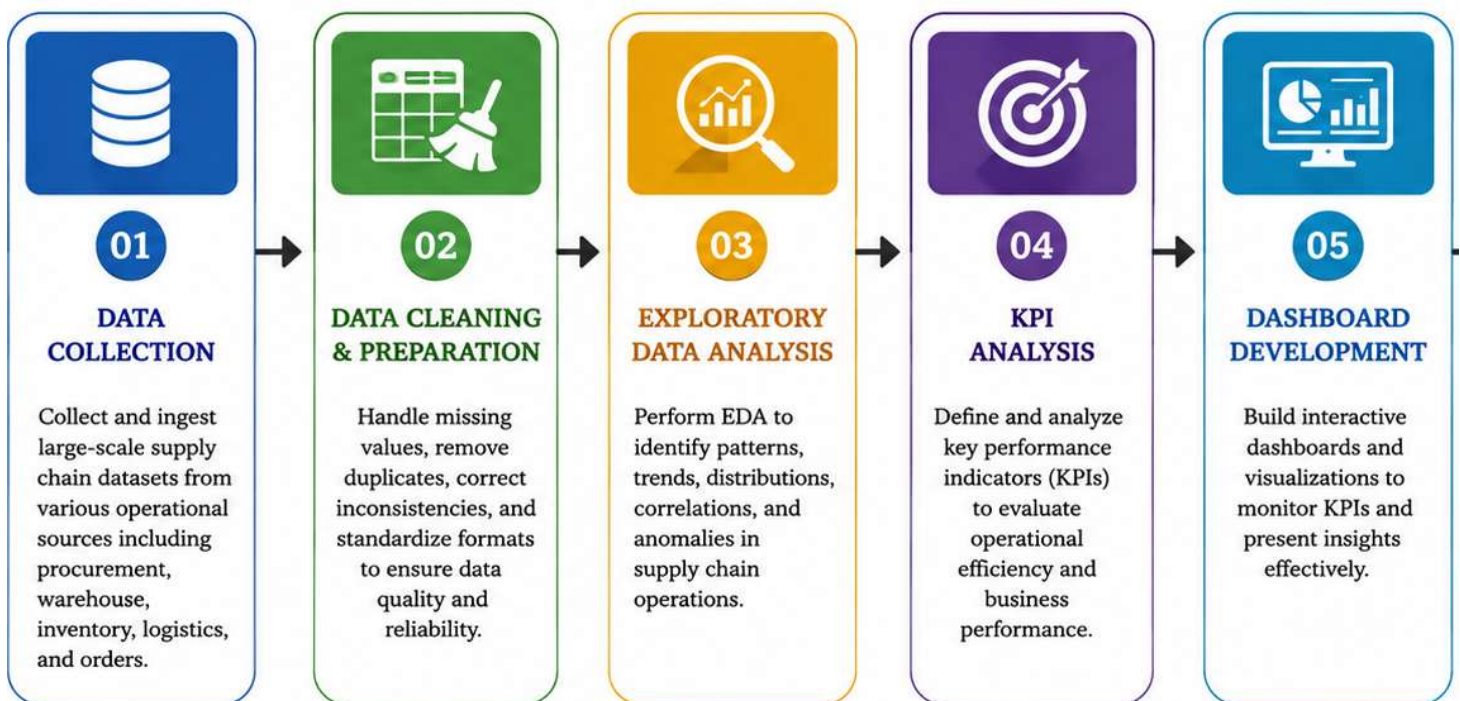
The methodology followed in Phase 1 was designed to systematically analyze historical supply chain datasets and extract meaningful insights through descriptive and diagnostic analytics techniques. The workflow began with collecting and ingesting large-scale datasets containing procurement, warehousing, logistics, inventory, and order-related information.

The first step involved **Data Cleaning and Preparation**, where missing values, duplicate records, and inconsistencies were identified and handled. Data formats were standardized and preprocessing techniques were applied to ensure data quality and reliability.

The second stage involved **Exploratory Data Analysis (EDA)** to understand patterns, trends, and anomalies within the supply chain data. Various analytical approaches were used to study relationships between variables and identify operational behavior.

Further analysis involved defining **Key Performance Indicators (KPIs)** such as Order Fulfillment Rate, Inventory Turnover, Lead Time Variability, and Cost-to-Serve to evaluate business performance.

Interactive dashboards were then developed for visualization and insight generation. Finally, **Root Cause Analysis** techniques were applied to identify major factors affecting operational efficiency and support future project phases.



DATA CLEANING AND PREPARATION

Data cleaning and preparation formed the initial stage of Phase 1 and played a critical role in ensuring the quality and reliability of the supply chain dataset. The collected dataset contained more than **180,000 records and multiple attributes** related to procurement, inventory, logistics, customers, products, and order operations. Since real-world datasets often contain inconsistencies and incomplete information, preprocessing was necessary before performing analytical tasks.

The dataset was loaded into the Python environment and examined to understand its structure, dimensions, and attribute details. Missing values across different fields were identified and analyzed. Columns containing excessive missing values and irrelevant information were handled appropriately. Duplicate records were detected and removed to improve data consistency.

Data preprocessing activities also included standardization of formats, handling inconsistent categorical values, trimming unwanted spaces, and verifying schema consistency. Outlier detection methods were considered to identify unusual observations that could affect analytical results. These cleaning procedures improved overall data quality and prepared the dataset for exploratory analysis.

The following preprocessing activities were performed:

- Dataset loading and structure inspection
- Missing value identification and handling
- Removal of duplicate records
- Detection of inconsistent values
- Standardization of formats and attributes
- Data validation and preprocessing

These preprocessing steps established a clean and structured dataset suitable for further analysis and dashboard development.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics and behavior of the supply chain dataset and identify meaningful patterns from historical data. EDA plays an important role in transforming raw information into actionable insights by summarizing key characteristics and relationships within the dataset.

The analysis was carried out using statistical techniques and visualization methods to examine trends, distributions, and operational behavior across various supply chain components such as procurement, inventory, warehousing, logistics, and order management. Both numerical and categorical attributes were analyzed to understand data variation and detect unusual patterns.

Different forms of analysis including **univariate, bivariate, and multivariate analysis** were conducted. Distribution analysis was used to study the behavior of variables, while correlation analysis helped identify relationships between different operational parameters. Seasonal patterns and anomalies affecting supply chain performance were also examined.

The EDA process included:

- Univariate analysis of individual variables
- Bivariate analysis to identify relationships between attributes
- Multivariate analysis for combined pattern identification
- Distribution and trend analysis
- Correlation analysis and anomaly detection
- Identification of seasonal and operational patterns

The insights generated through EDA provided a better understanding of supply chain behavior and supported KPI definition and dashboard development activities.

KPI FRAMEWORK & BUSINESS MAPPING

Key Performance Indicators (KPIs) were defined and analyzed to measure supply chain performance and evaluate operational efficiency. KPIs provide measurable values that help organizations monitor performance, identify problem areas, and support data-driven decision-making. In this phase, important supply chain metrics were selected based on business requirements and operational objectives.

The selected KPIs focused on different areas including inventory management, order processing, delivery performance, and cost optimization. Each KPI was mapped to specific business questions to understand how operational activities influence overall supply chain performance.

The KPI analysis helped identify patterns and assess operational effectiveness across multiple supply chain functions. Performance metrics were further utilized for dashboard visualization and insight generation.

The following KPIs were considered during analysis:

- **Order Fulfillment Rate** – Measures successful completion of customer orders.
- **Inventory Turnover** – Evaluates inventory movement efficiency.
- **Lead Time Variability** – Measures variation in delivery or processing time.
- **Perfect Order Rate** – Indicates percentage of error-free deliveries.
- **Cost-to-Serve** – Measures operational costs associated with serving customers.

These KPIs established a structured framework for analyzing supply chain performance and answering important business questions.

DASHBOARD DEVELOPMENT

Dashboard development was carried out to transform analytical findings into interactive visual representations that support effective decision-making. Dashboards provide a centralized platform for monitoring supply chain performance and understanding operational behavior through graphical insights. The objective of this stage was to present complex supply chain information in a clear and user-friendly format.

Interactive dashboards were designed to visualize procurement, inventory, logistics, supplier performance, and order-related metrics. Various charts, graphs, and visual indicators were used to summarize business performance and enable better interpretation of large-scale datasets. Drill-down functionality was considered to provide deeper analysis of specific operational areas.

Visualization tool **Power BI** was utilized to create dashboards capable of presenting supply chain insights effectively.

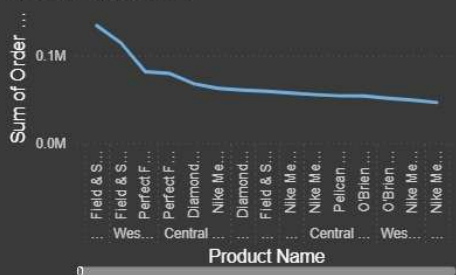
The dashboard included visual representations of:

- Regional performance analysis
- Supplier reliability scores
- Demand trend visualization
- Inventory health monitoring
- KPI performance indicators
- Supply chain operational insights

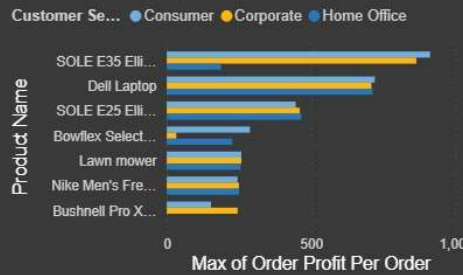
The developed dashboard provided a structured view of business performance and helped identify important trends and operational patterns within the supply chain system.

Vaibhav Yedla's Sales Analysis Report

Sum of Order Profit Per Order by Order Region and Product Name



Max of Order Profit Per Order by Product Name and Customer Segment



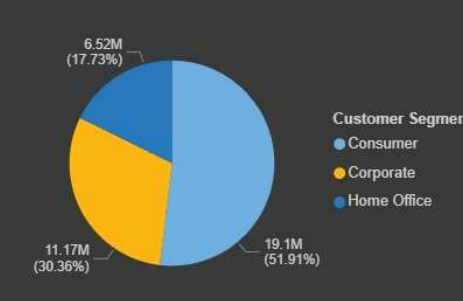
Sum of Late_delivery_risk by Delivery Status and Order Region



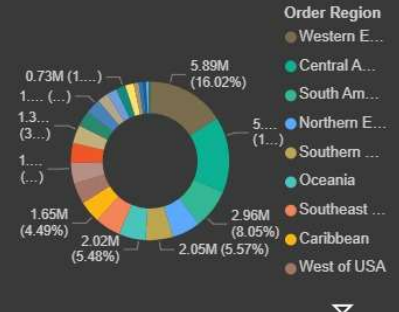
Sum of Sales and Sum of Order Profit Per Order by Category Name



Sum of Sales by Customer Segment



Sum of Sales by Order Region



FINDINGS

The analysis of the supply chain dataset provided several meaningful insights regarding operational performance and historical trends. Data preprocessing improved dataset quality and ensured reliable analysis. Exploratory analysis helped identify patterns related to inventory movement, logistics activities, order processing, and customer behavior.

The major findings obtained during Phase 1 include:

- Large-scale datasets required preprocessing before analysis.
- Missing values and duplicate records affected data quality.
- Supply chain performance varied across different operational parameters.
- KPI analysis helped understand business efficiency and operational trends.
- Dashboard visualizations improved interpretation of analytical results.

The findings established a strong foundation for future predictive analysis and optimization activities.

CONCLUSION

Phase 1 successfully focused on descriptive and diagnostic analysis of historical supply chain data. Activities including data cleaning, preprocessing, exploratory analysis, KPI identification, and dashboard visualization helped derive meaningful business insights from large-scale datasets.

The analysis process improved understanding of operational patterns and supported data-driven decision-making. The outcomes of this phase provide a structured analytical foundation for upcoming phases involving predictive modeling, optimization techniques, and intelligent decision-support systems.

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